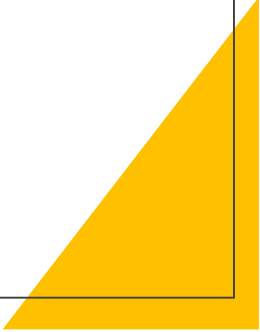


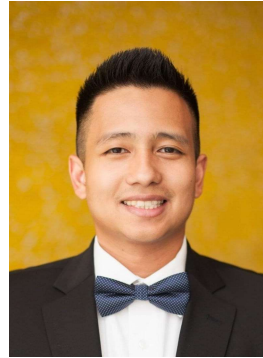
Introduction to Musculoskeletal Care

Bronwyn Sing, DO, MS, C-NMM

ISMSN

Fall 2025

A yellow triangular graphic is located in the bottom right corner of the slide, pointing towards the center.



Welcome!

Bronwyn Sing, DO (Clinical Sciences) FM/OMM

Kyaw Lin, DO (Clinical Sciences) PM&R

Richard Aptaker, DO (Clinical Sciences adjunct faculty)
PM&R

Learning Objectives:

After this session, the OMS II student will be able to:



Define essential MSK terminology and structures



Describe the pathophysiology of muscle, tendon, and ligament injury and related them to findings on physical exam.



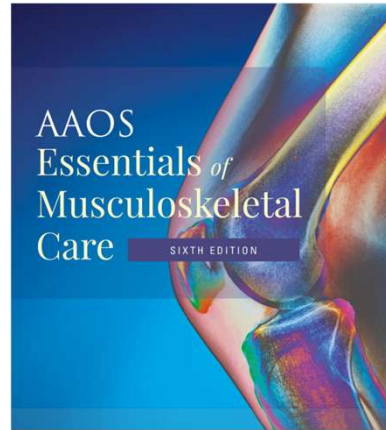
Distinguish acute from chronic soft-tissue pathology



Apply these principles to clinical cases

Resources

- Textbook: Essentials of Musculoskeletal medicine, 6th ed.
- Online Med Ed, Trulearn, Lecturio
- Physical Exam: Touro MSK videos
- Office Hours TBD, Clinical Sciences fellows



Dr. Lin and I have reviewed COMLEX blueprint

Tips on Success in this Module



Know your anatomy!



Practice on yourself, a housemate, anyone willing to help!



Come prepared to the labs, ask questions, study to understand the mechanisms, not just to remember for a test

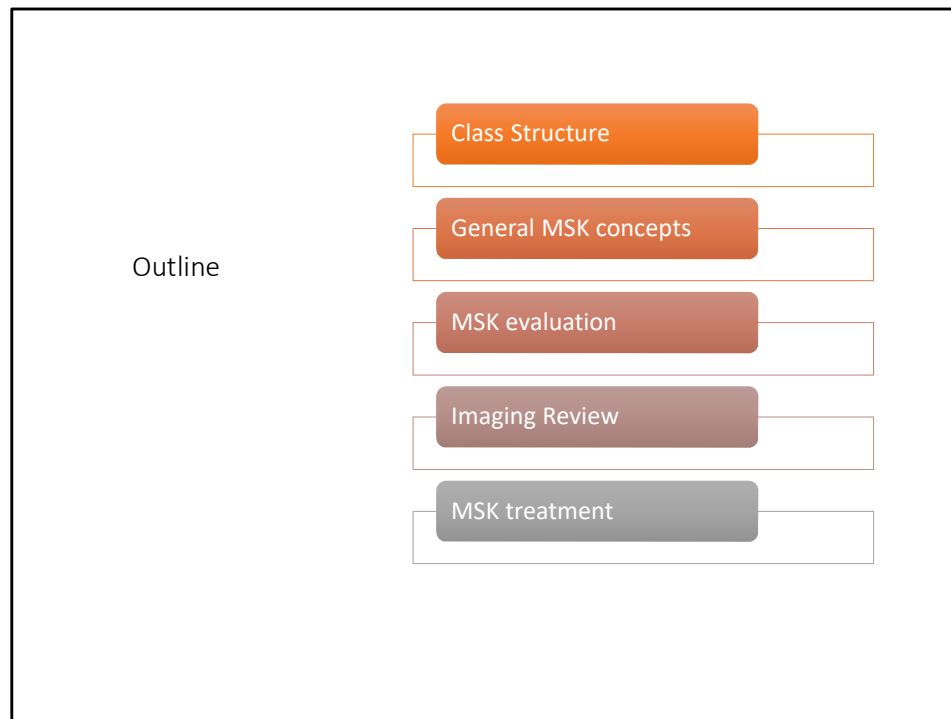


Interweaving: Revisit and Integrate the material (Questions in ISMSN will use OD3 material for testing purposes)

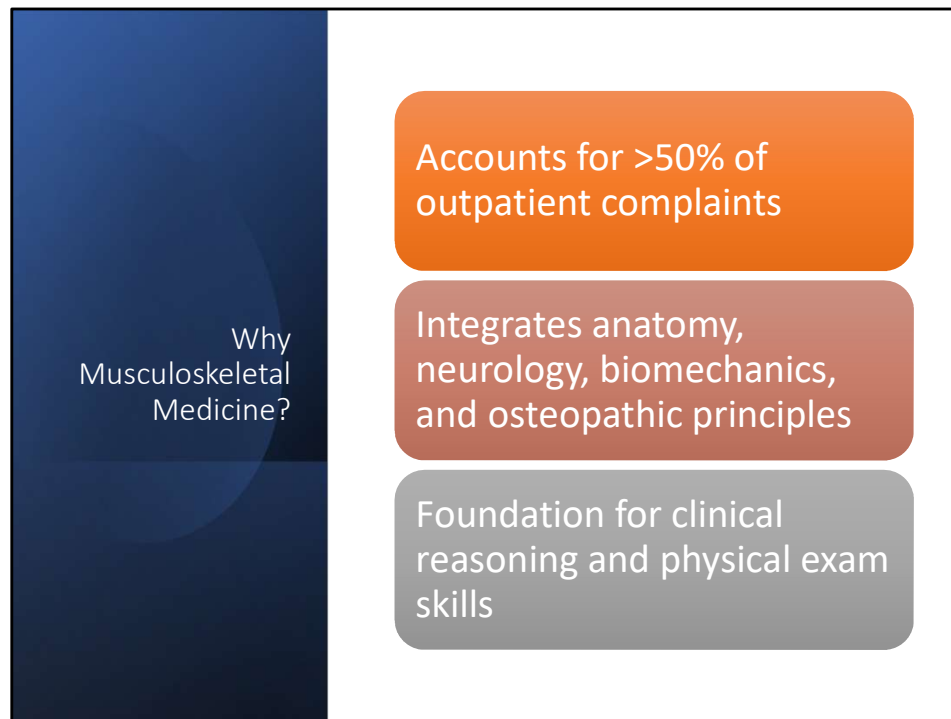


Have fun with learning this material!

This course is going to give you the tools to have systematic approach, know what is urgent and what takes time to heal. MSK is a great topic to integrate your OMM skills. Where you can differentiate yourself as a medical student/physician. After you assess and know what you want. You are different from PT and trainers. Patients do well with OMM and you will be able to assess, diagnose, and treat with OMM if appropriate.



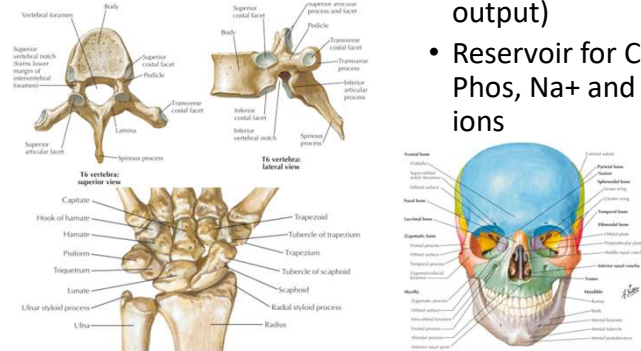
Anatomy, IS-MSN, OD3, OPP3...presented in a way that will help you integrate all the information



Also many of us personally experience MSK issues as well as family and friends. These are skills you will revisit many times in your career!

Bones: structure and function

- Load bearing
 - Mobility
 - Protection
-
- Hematopoiesis
 - Bones are vascular (get about 10% cardiac output)
 - Reservoir for Ca^{+} , Mag, Phos, Na^{+} and other ions



Long Bones- provide mobility, can sustain a high load of weight. Ex) femur, tibia.

Short Bones- provide support and stability, less mobility ex) carpal bones

Flat bones- provide protection of other structures ex) frontal bone, skull, sternum.

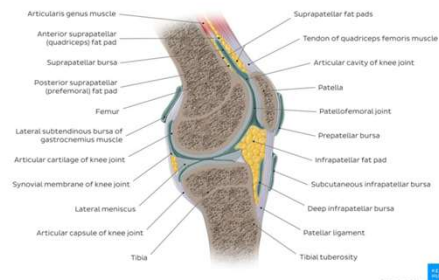
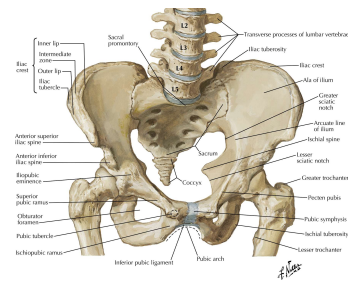
Irregular Bones- don't fall into a long or short bone category.

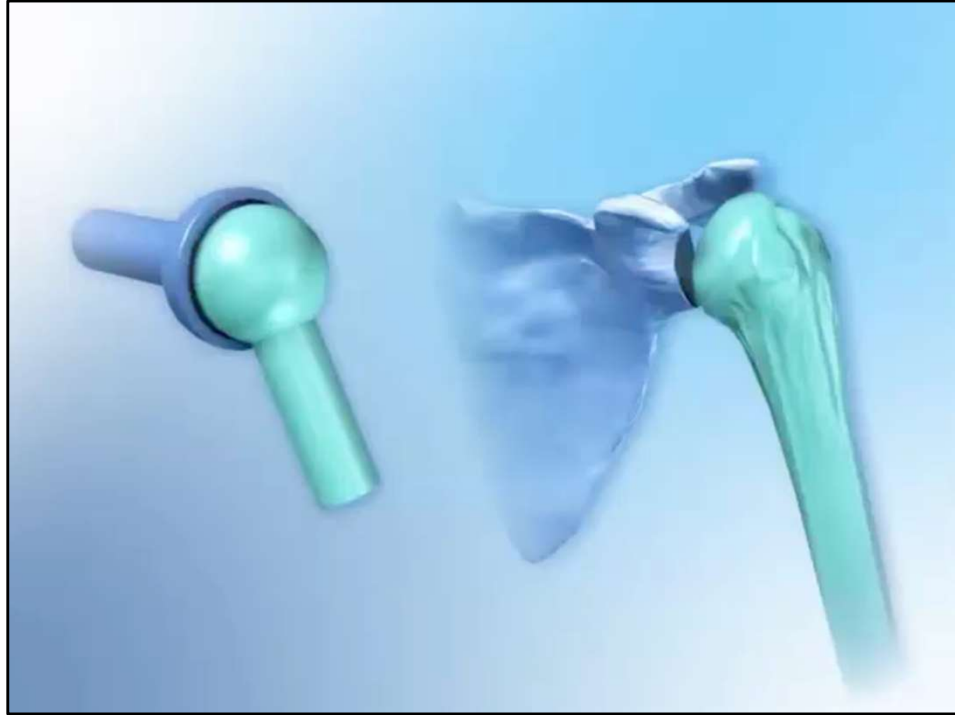
Pneumatic-contain hollow air filled cells ex) sphenoid

Sesamoid bones- thought to aid in balance. Ex) under great toe, patella.

Joints

- **Fibrous**
 - Little movement
 - Ex) cranial sutures, distal tib fib
- **Cartilaginous**
 - Limited movement
 - Ex) pubic symphysis
- **Synovial**
 - Freely movable, has synovial fluid
 - Ex) shoulder, knee, hip

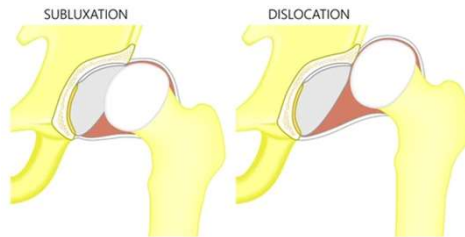




Functional Examples

- Hip (Ball-and-Socket):** Deep acetabulum → stability for weight-bearing, yet allows multidirectional motion.
- Shoulder (Ball-and-Socket):** Shallow glenoid → large ROM, less stability → prone to dislocation.
- Thumb (Saddle Joint):** Permits **opposition**, essential for grasp and fine motor control.
- Temporomandibular Joint (TMJ):** Modified hinge/condyloid joint → allows opening/closing, protrusion/retraction, side-to-side motion.
- Spine (Facet Joints):** Plane synovial joints → guide and limit motion, provide load sharing and stability.

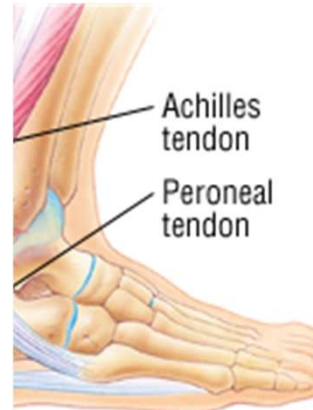
Subluxation	Dislocation
A minor or incomplete dislocation. Joint surfaces still touch but are not properly aligned.	An injury that causes two adjoining bones to separate from one another fully



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MSK Tissue: Structure and Function

- Muscle: movement, posture, heat generation
- Tendon: connects muscle to bone, transmits force
- Ligament: connects bone to bone, provides stability
- Bursa: reduces friction



Ligaments- connect bone to bone, Provide stability and provide limitations to range of motion. **Ligaments** are tough fibrous cords composed of connective tissue that contains both collagen and elastic fibers. The elastic fibers allow the **ligaments** to stretch to some extent. **Ligaments** surround joints and bind them together. They help strengthen and stabilize joints, permitting movement only in certain directions. **Ligaments** also connect one bone to another (such as inside the knee).- MERCK MANUAL

When Injured it can tear partially or fully, ligaments can be damaged and you can feel laxity.

Bursa: Fluid-filled, saclike cavities are known as bursae. These are located where muscles and tendons move over bony joint areas. Their function is to reduce friction caused by muscles and tendons moving against skin and bones, as well as to facilitate movement.

Bursitis: Bursa become inflamed. Creating localized pain that is exacerbated by specific movements and relieved by rest. Palpation over the bursa will create

tenderness.



Key MSK Terminology

Sprain – Ligament injury

Strain – Muscle/tendon injury

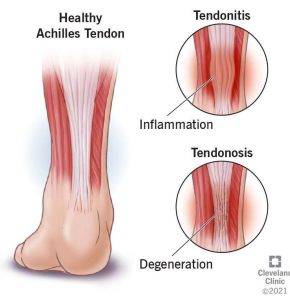
Tendinopathy – General tendon disorder that encompasses tendinosis, tendinitis.

Bursitis – Bursa inflammation

Muscle Pathology

- Strain – overstretch/overload → pain, weakness
- Spasticity – UMN lesion, velocity-dependent tone
- Rigidity – basal ganglia lesion, lead-pipe tone

Tendinopathies



- **Tendinitis** – acute inflammation (i.e .rotator cuff, Achilles)
- **Tendinosis** – chronic degeneration, disorganized fibers, scarring (i.e. lateral epicondyle)
- **Tenosynovitis**- Inflammation of a tendon and its enveloping sheath. (i.e. de quervains syndrome)
- **Enthesitis** – insertional inflammation, painful on palpation (i.e psoriatic arthritis)

Image: <https://my.clevelandclinic.org/health/diseases/22289-tendinopathy>

Ligament Pathology

Provides joint stability

Poor vascularity → slow healing

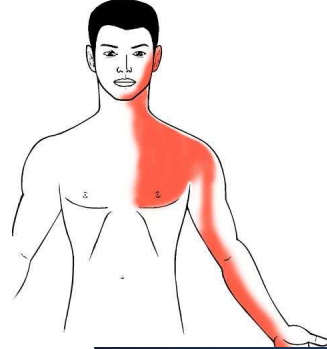
Grades I–III: stretch → partial tear → full tear

Example: ACL sprain

Pain Patterns

- Local – pain at site of injury, pain reproduced (i.e. muscle strain, bony fracture)
- Referred – distant but related pain (cardiac → shoulder)
- Radicular – pain that travels along nerve root (i.e. “sciatica” or lumbar radiculopathy)

Heart-Related Pain



What causes pathology in these structures?

Overuse: causes inflammation, wear and tear, degeneration (tendons, bone and cartilage)

Inflammation: Rheumatologic, infectious

Bony changes (i.e. osteoporosis, arthritis)

Trauma (reported or not!)

Malignancy

Primary malignancy

Metastatic bone disease: breast, prostate, lung, kidney, thyroid

Suspected abuse, does the story/mechanism make sense, does the physical exam make sense? Keep asking, we are mandatory reporters!

Case 1: Acute Ligament Injury



24-year-old soccer player feels 'pop' in knee after jump.



Likely structure: ligament (ACL)



Expected findings: swelling, instability



What if this was a muscle strain?

Pain with ROM of muscle when engaged, +TTP at muscle belly.

Ecchymosis?

Swelling?

Case 2: Tendon Pain

- 42-year-old office worker with lateral elbow pain for 6 months.
 - Likely diagnosis: tendinosis
 - Findings: tenderness distal to lateral epicondyle
-
- When would you suspect Tendonitis?

History & Exam Highlights

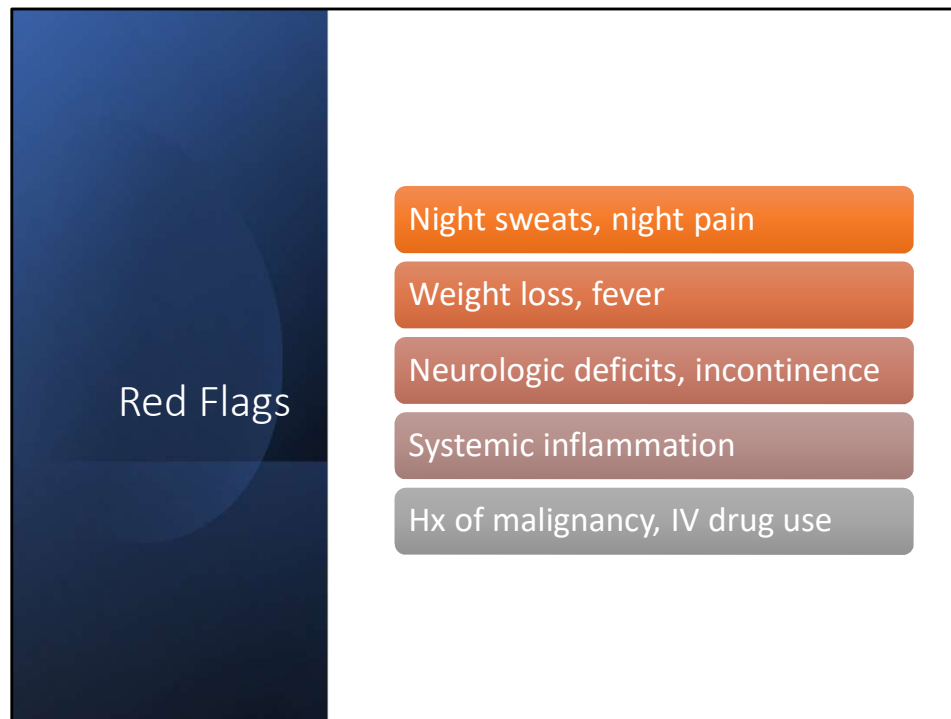
Use OLD CARTS + mechanism

Functional questions (walking, lifting, job, ADLs, dexterity)

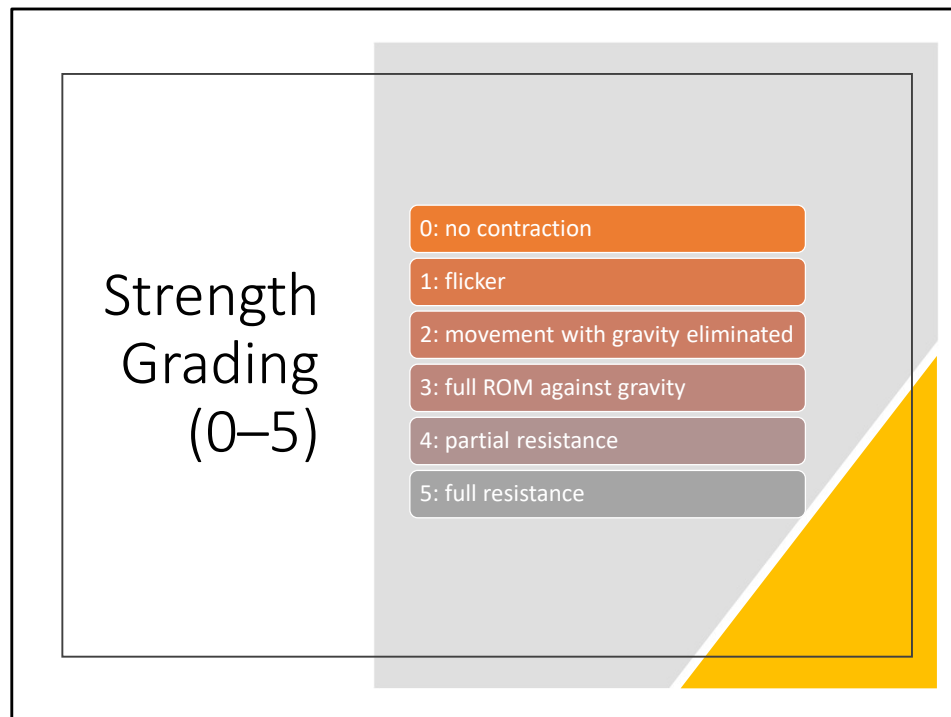
Compare sides (most will have 2!)

Inspect → Palpate → ROM → Strength → Neurovascular

Then special tests!



This is review but always make sure you have your suspicion high

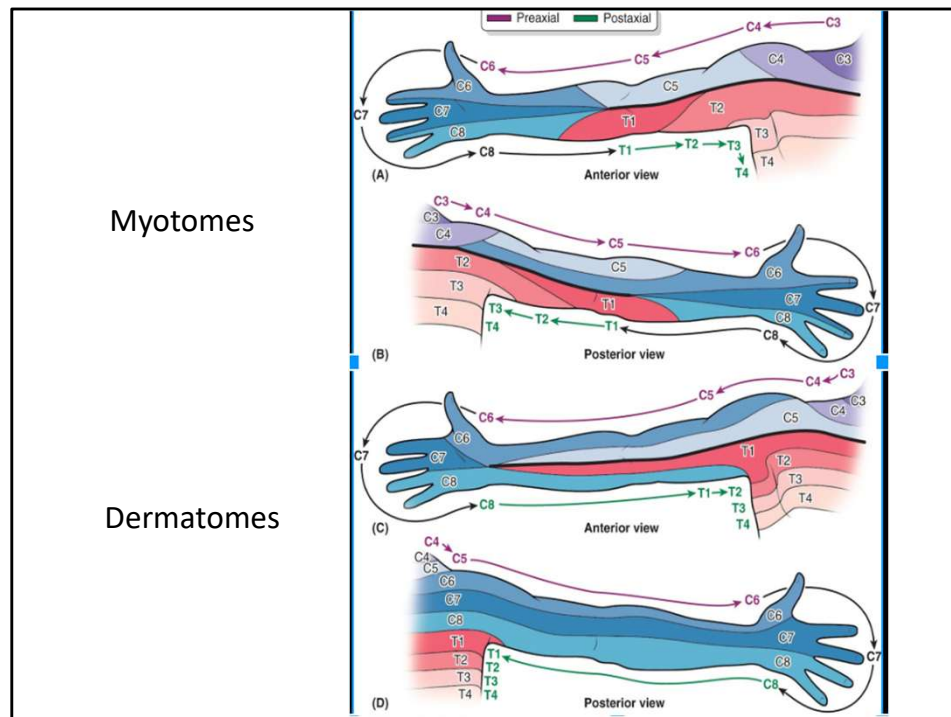


We pay attention to weakness, new and old!



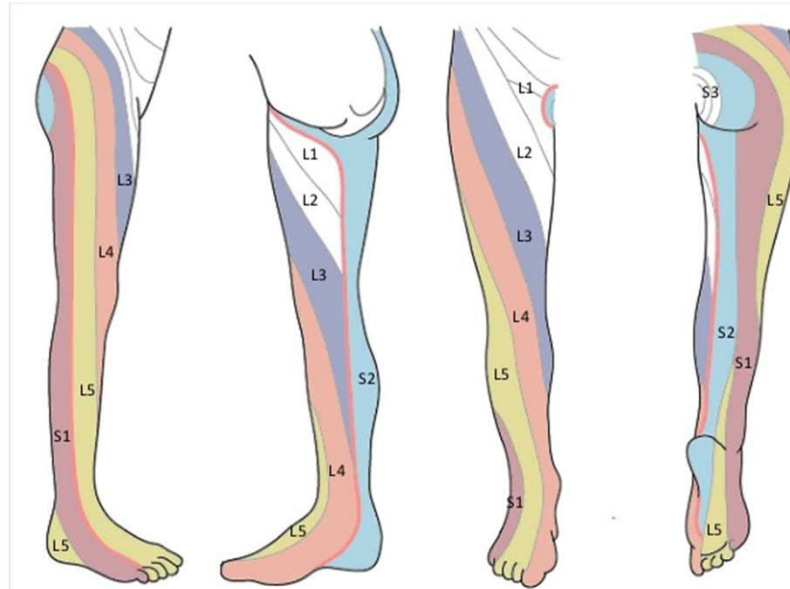
Neuro Exam

- Sensation
- Strength
- Reflexes



Dermatomes coming back to haunt you, this is essential in your MSK evaluation.

Lower Extremity Dermatomes



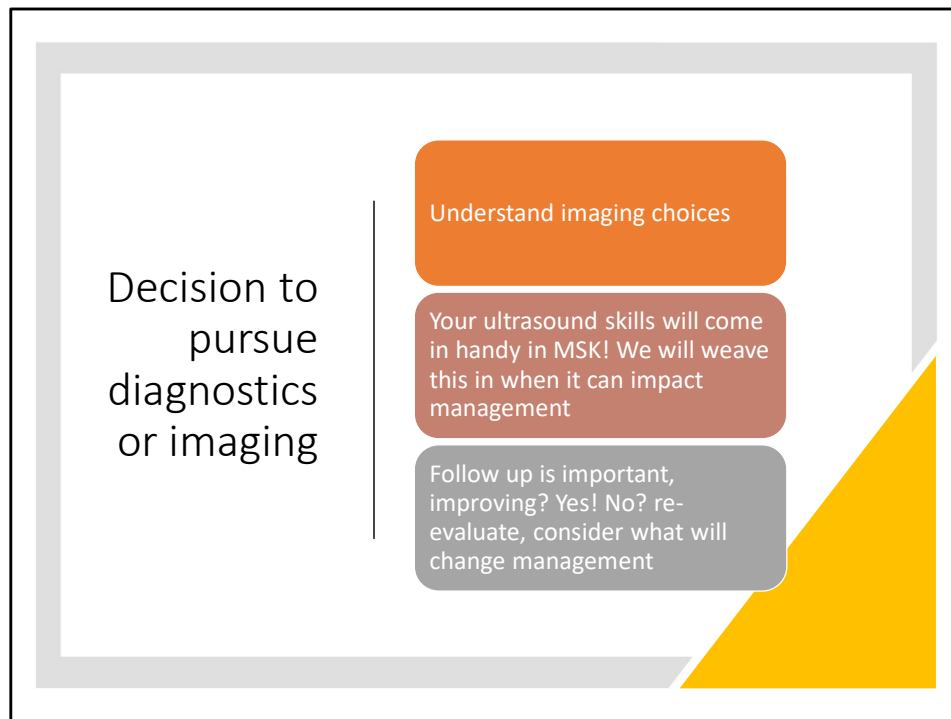
Reflexes

Grade	Description
0	Absent
1+ or +	Hypoactive
2+ or ++	"Normal"
3+ or +++	Hyperactive without clonus
4+ or ++++	Hyperactive with clonus

Osteopathic Integration

- Evaluate for somatic dysfunction
- Treat safely when appropriate
- OMM can enhance recovery and function

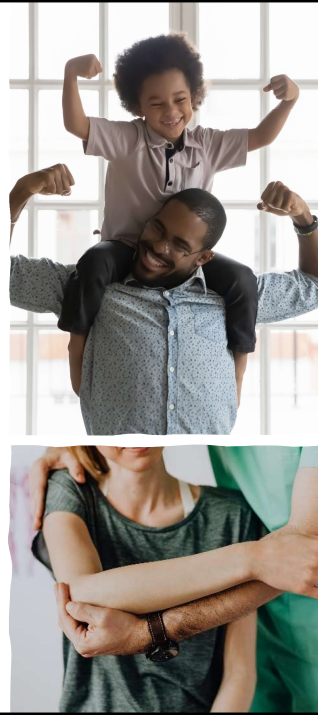




Dr. Grammy has created an in-depth guide for your MSK related radiology review.

Basic Management Principles

- The body heals itself — support it!
- RICE, relative rest
- NSAIDs short term
- Rehabilitation and conditioning



Rest, Ice, compression, elevation
If you don't use it, you lose it.
Support ADLs, mobility



Short term or long term, mobility provides more independence and access

Medications

- Acetaminophen- pain and anti pyretic
- Nonsteroid Anti-inflammatory Drugs (NSAIDs)
 - Use short term
 - Treats pain and inflammation
- Muscle relaxers
- Opioids
 - Use very short term, specific use of this medication.



Acetaminophen- processed hepatically.

NSAIDS- ibuprofen, naproxen etc- processed renally. High risk of GI ulcers and GI bleeding.

Muscle relaxants (cyclobenzaprine, baclofen)

Opiate medications- specific indications

DMARDS (Disease-modifying antirheumatic drugs)- used in Autoimmune systemic disease, more to come in Rheumatology.

Splinting & bracing



Short Arm Cast



Long Arm Cast



Arm Cylinder Cast

Provide temporary support to the ligament, muscle, or bone that is affected. Recommend Short term use, as long term it will promote muscle dependence and weakness. Goal is to allow for the structure to heal and then removing the brace will allow the structures to slowly strengthen and return back its previous function.

Take-Home Points

Learn the language of MSK medicine

Know tissue types and failure patterns

Mechanism + structure → pathology

Integrate osteopathic principles early

Resources

Essentials of
Musculoskeletal
Medicine, 6th ed.

OnlineMedEd,
Trulearn, Lecturio

Touro MSK
Physical Exam
Videos

COMBank
questions for
review